

Symmetric-Key Cryptosystem

 $\mathcal{P}$ $\mathcal{K}$ $\mathcal{C}$ $\mathcal{E}$ $\mathcal{D}$

$$\forall P \in \mathcal{P}, K \in \mathcal{K}, \\ \mathcal{D}_K(\mathcal{E}_K(P)) = P$$